Feature Importance Analysis

Machine Learning-Based Analysis of Household Expenditure in Urban Maharashtra

1. Methodology

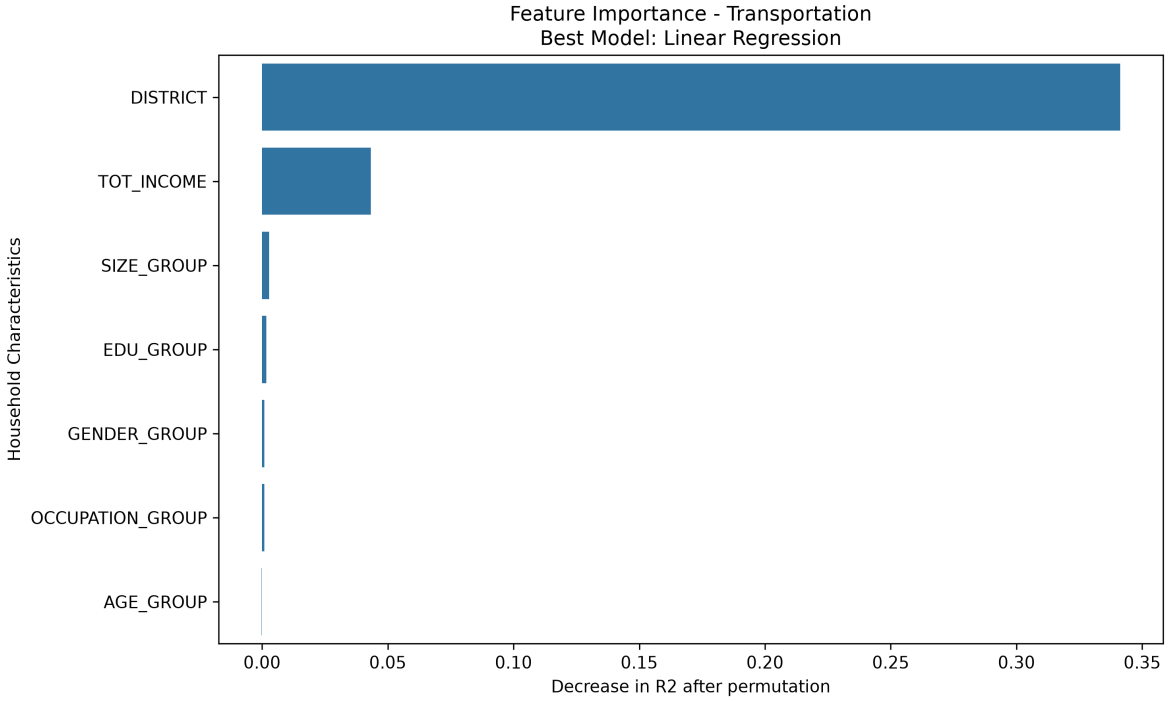
Permutation importance was used to evaluate the predictive contribution of household characteristics. The method measures the reduction in model R2 score after randomly shuffling each feature. The analysis uses the best-performing model selected separately for each expenditure target.

2. Feature Importance Results

Transportation Expenditure

Selected model: Linear Regression

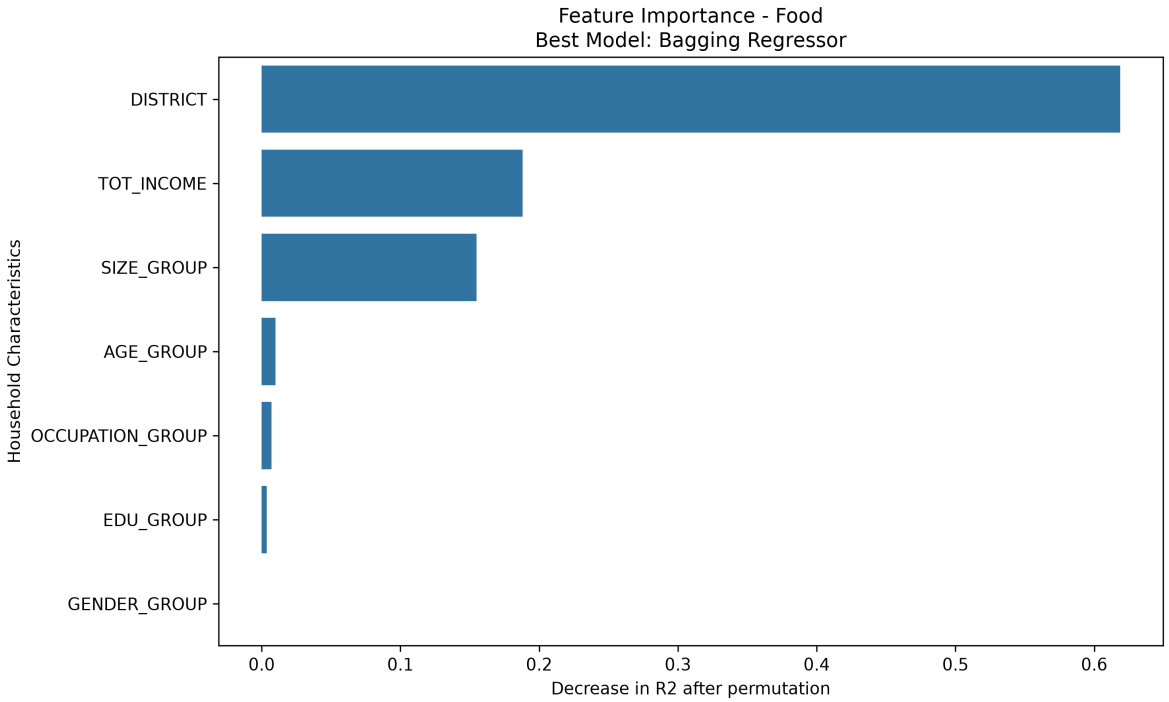
Feature	Importance	Std. Dev.
DISTRICT	0.34124	0.01512
TOT_INCOME	0.04325	0.00523
SIZE_GROUP	0.00282	0.00115
EDU_GROUP	0.00172	0.00162
GENDER_GROUP	0.00099	0.00110
OCCUPATION_GROUP	0.00091	0.00056
AGE_GROUP	-0.00019	0.00078



Food Expenditure

Selected model: Bagging Regressor

Feature	Importance	Std. Dev.
DISTRICT	0.61869	0.03002
TOT_INCOME	0.18806	0.01172
SIZE_GROUP	0.15471	0.01242
AGE_GROUP	0.01001	0.00377
OCCUPATION_GROUP	0.00697	0.00398
EDU_GROUP	0.00369	0.00497
GENDER_GROUP	-0.00007	0.00555



3. Interpretation and Limitations

Features with larger positive permutation importance values contribute more to the selected model's predictive performance. Near-zero values indicate limited additional predictive contribution under this evaluation. Negative values can occur because of sampling variability or model behaviour.

Feature importance indicates predictive relevance and does not establish a causal relationship. Correlated household characteristics can also share or obscure predictive importance.